LEARNER GUIDE

FACULTY	NATURAL AND APPLIED SCIENCES
QUALIFICATION	ADVANCED DIPLOMA ICT (APPL. DEVELOPMENT)
QUALIFICATION CODE	
MODULE	DEVELOPMENT SOFTWARE
MODULE CODE	NDEV74116
COMPILED BY	DR COLIN CHIBAYA

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1. GENERAL INFORMATION

1.1 Welcome and Introduction

On behalf of Sol Plaatje University, we welcome you to the Advanced Diploma in ICT programme with a specialization in Applications Development, and more specifically to Development Software as a module. This module is mainly intended to logically continue from the programming modules offered in the diploma programme to more advanced programming topics. The content will include the use of Java (NetBeans) advanced features. Detailed module information is given in subsequent sections of this learner guide, showing you the number of lectures, practical and tutorial periods per week, and providing information about additional time you will be expected to spend on module-related work. Your responsibilities about participation and attendance are stated and an overview is provided of the learning and teaching approaches. Comprehensive assessment details for the module are also given, including the materials that will be used.

This learner guide also provides you with guidelines on what to expect from the University and what the University, in turn, expects from you regarding this module. The module is divided into five sections, all of which collectively contribute towards preparing you for the challenges faced in commercial software development environments.

In section 1, the module recaps on the fundamental principles of programming in Java. Key Java programming concepts are discussed, together with the main programming tools and constructs. In revisiting known sequential, conditional, and looping structures, emphasis will be on problem solving using the IF, switch, FOR, DO, and WHILE statements. Most importantly, Object Oriented Programming is visited, where and informed understanding of the concepts of classes, objects, constructors, polymorphism, overloading, passing of parameters, data encapsulation - public and private data, methods access modes is sought. Students should be able to work with multiple classes.

In Section 2, the module dwells on data management, emphasizing the different data structures required in handling data in memory. Precisely, the concepts of recursion, 1D arrays, 2D, 3D, 4D and nD arrays. The concepts of stacks, queues, and double linked lists are also introduced. Operations on these data structures are discussed, including searching, sorting, inserting, and deleting algorithms.

Section 3 extends the concept of data handling in memory to external data management using DB and text files. Key is students' understanding of CRUD operation on data held in DB or text files. Emphasis is on students' understanding of DDL and DML procedures. Aspects of data visualization are introduced, emphasizing the development of graphic primitives.

In section 4, the course dwells on the concept of graphs as a data structure. The concept of graphs, nodes, paths, edges, connections, completeness, and weighting. Types of graphs and representation. Operations and implementation, are the key focus areas

Lastly, section 5 presents a few selected sets of advanced algorithms including the Divide and conquer algorithms, Greedy algorithms, and Dynamic algorithms.

The Advanced Diploma qualification is primarily professional and industry specific. The knowledge students acquire emphasizes on general principles and application, developing the graduates who can demonstrate focused knowledge and skills in applications development.

1.2 Lecturer & Tutor Information

	Name	**E-mail address**	**Consultation**
Lecturer	Dr. Chibaya, Colin	colin.chibaya@spu.ac.za	

1.3 General Course Information

Credit Value	**NQF Level**	**Field**	**Sub-Field**
16 Credits	7	Information Technology	Application development

1.3.1 Formal Contact Time – attendance at all classes is compulsory. The lecturer will administer unannounced quizzes which contribute towards the final CASS mark every now and again to curb attendance issues.

Theory	4 x 45-minute periods per week – preferably two double periods per week
Practical	2 x 45-minute periods per week - preferably one double period per week
	preferably one double period per week

1.3.2 Non-Contact Time

Students are expected to spend at least six hours in a week self-studying and completing work outside of the formal contact time and tutorial sessions.

1.3.3 Notional Learning Hours

Activity	**Code**	**Hours**
Lectures	L	30
Tutorials	T	15
Practical Work	PW	20
Laboratory Work	LW	-
Independent self-study of standard texts and references (study guides, books, journal articles)	IS1	15
Independent self-study of specially prepared materials (case studies, multi-media, etc.)	IS 2	20
Assessments (Summative & Formative)	A	20
ICT (e.g. Moodle, Communications Lab, etc.)	ICT	10
Presentations (by students)	P	10
Research	R	30
Service Learning	SL	-
Others	O	-
Total		160

1.3.4 Experiential Training/Service Learning/Work Integrated Learning (WIL)

There is currently no WIL component in this course.

1.4 Learning and Teaching Strategies

The University has adopted a smart based student-centred, interactive teaching and assessment strategy applicable.

1.5 Assessment Criteria

1.5.1 Formative Assessment

Formative assessment will be used to inform both the student and the lecturer regarding progress. It will be allocated marks. We use this type of assessment to allow the student and lecturer to jointly monitor progress in achieving the module goals and specific module outcomes. It will take the form of self-review, peer review, and lecturer reviews, and should be seen by the student as a valuable learning aid. It will also be used to provide valuable feedback to the lecturer regarding his/her instruction techniques, topics that require revision or additional attention. Students will be asked to provide comprehensive evaluations of the course and lecturer at least once during the course cycle. All formative assessments will count towards the semester mark as indicated in the table in section 1.5.2 below.

1.5.2 Summative Assessment

This type of assessment is used to formally measure the learning of particular skills, knowledge and understanding and is associated with a recorded mark. All summative assessments will count towards the semester mark as indicated in the table below:

Assessment	Description (Scope)	Weight
Formative Assessment 1 (Assignment)	All topics	20
Formative Assessment 2 (Class Activities)		10
Summative 1 (Major Test 1)	Topics 1, 2, 3	35
Summative 2 (Major Test 2)	Topics 4, 5	35
Total		**100**

Please Note: A student must score at least 50% for the final mark (calculated as shown above) to pass the course.

Medical certificates regarding missed tests and/or assignments, etc. must be submitted to the lecturer or site department secretary within 5 working days of the test date or due date. Medical certificates not received within 4 days will not be considered – unless the student was hospitalized. If a medical certificate is accepted, an assessment would be prepared for the student. Complaints or queries about assessment marks must be raised with the lecturer concerned within 7 days of the assessment being returned to the students.

1.5.3 Examination

There will be a 3 hour written examination at the end of the semester. However, there will be a subminimum of 40% required to pass an examination. If a student scores below 40% in the examination, he/she does not qualify for a final mark and does not pass the course.

1.5.4 Final Mark Calculation - Opportunity Final Examination

The Final Mark is calculated as follows:

Semester mark	60%
Examination	40%
Total	100%

A student must score at least 50% for the final mark (calculated as shown above) in order to pass the course. A student may be allowed to write a second opportunity examination if missed or failed the first opportunity examination and a Final Mark would be calculated using the same criteria used in first opportunity.

1.5.5 Re-marking, absence from exams due to illness, special examinations

See General Prospectus for rules pertaining to these.

1.6 Learning Materials and Stationery Requirements

1.6.1 Prescribed Textbook(s)

- Liang, Y,D. "Introduction to JAVA programming". Pearson International Edition. Seventh edition. ISBN: 10: 0- 13- 605966 – X.13: 978-0-13-605966 – 0. 2009.
- Liang, Y,D. "Introduction to JAVA programming and Data Structures". Pearson International Edition. Eleventh edition. ISBN: 10: 1- 292- 22187 – 9
  13: 978-1-292-22187 – 8. 2018.

1.6.2 The following books are also recommended reading:

- Any other Java programming textbook/resource can be used in this module

1.7 Plagiarism

Plagiarism should be always avoided, whenever a student uses an author's work without recognizing it and /or cutting and pasting it (not stating facts/thought in own words). Plagiarized work will not be accepted for marking and students may be subjected to disciplinary action.

2. TECHNICAL COMPONENT

2.1 Admission Requirements and Pre-Requisites

Refer to University General Prospectus

2.2 Course Exit Level Outcomes

This module is intended to provide students with exposure to advanced software development management skills. This is a compulsory module that currently supports specializations in Applications Development and Multimedia Applications.

2.3 Moderation

All summative assessment question papers and memoranda for Development software are externally moderated at the end of the semester.

2.4 Articulation

Completion of this module offers vertical articulation to the Post Graduate Diploma in Information Technology (Application Development) and is portable to other institutions offering the same Advanced Diploma.

3. LEARNING COMPONENT

3.1 Topics and Specific Outcomes

Topic	Competencies	Learning Outcomes – students will be able to:
1. Fundamental principles of programming (A recap)	Students will be required to demonstrate an informed understanding of the basic principles of programming in Java – on the NetBeans platform. Important topics include an understanding of the program structure, concepts of variables and data types, data input and output, arithmetic, relational, and logic operators, linear programs, Conditions – the IF and switch statements, Loops – FOR, DO, and WHILE statements, Methods and string handling. We highlight the importance of the OOP while defining classes and objects, sub-classes, constructors, data hiding, overloading, polymorphism	• Demonstrate an informed understanding of the java programming and problem-solving using OOP views.

2. Data Management in the primary memory	The topic on data management in the primary memory will be introduced emphasizing topics on recursion, 1D, 2D, 3D, 4D and nD arrays, stacks, queues, double linked lists, as well as binary trees.	• implementation of data structures and related algorithms • Solving problems
3. Data Management in the secondary memory	This topic explains the processes involved in working with data stored in the secondary memory in repositories such as text files and databases. Methods for data retention in external devices are key, including DDL and DML operations in SQL	• implementation of algorithms to solve related problems. • Achieve CRUD operations
4. Graphs as a data structure	The concept of graphs, nodes, paths, edges, connections, completeness, and weighting. Types of graphs and representation. Operations and implementation.	• Understand how to apply this knowledge in real life situations
5. Advanced algorithms	A few selected sets of advanced algorithms are considered, including the Divide and conquer algorithms, Greedy algorithms, and Dynamic algorithms	• Understand how to apply this knowledge in real life situations

3.2 Work Schedule: Development Software (16 CREDITS)

Notes:

a) Refer to the table in 3.1 for the Resources needed.

b) Refer to 1.5 for the Learning and Teaching strategies that will be used.

c) Refer to 1.6 for Assessment information.

When possible, summative tests will take place every 4 or 5 weeks.

Below is a proposed work schedule which may change due to unforeseen circumstances.

Week	Specific Outcome	Date Completed	Comment
Week 1 and Week 3	- Topic 1		Students to complete a few lab tasks in which most concepts are implemented
Week 4 to Week 8	- Topic 2		Students to complete a few lab task in which most data structures are implemented
Week 9 And Week 10	- Topic 3		Students to complete a few lab tasks in which most concepts are coded
	Test 1		
Week 11	- Topic 4		Students to complete a few lab tasks in which most concepts are implemented

Week 12	- Topic 5		Students to complete a few lab tasks in which most concepts are implemented
	Test 2		
	Examination		

3.3 Critical cross-field outcomes

- Identify and creatively solve basic commercial/scientific problems within the context of each scenario given. You can achieve this by:
    - applying basic logic skills
    - applying basic business skills
    - applying programming principles
    - making informed and meaningful decisions
    - thinking creatively
    - considering alternative solutions
- Work effectively with others as a member of a team/group.
- Organize and manage yourself and your activities responsibly and effectively.
- Collect, analyze, organize, and critically evaluate information.
- Communicate effectively using language skills in the modes of oral and/or written persuasion.
- Use technology responsibly, effectively, and critically, showing responsibility towards the environment and health of others.

4 Roles, Responsibilities and Attendance Requirements

According to University Policy all lectures must be attended punctually and regularly. Attendance at all practical classes is especially important for students studying within the Department of Information Technology as the computer laboratory time is at a premium and skills gained in practical classes are fundamental for all IT courses. Registers will be taken at all classes and filed for future reference.

Participation is an important part of the class sessions. Participation means both attendance and engagement. Students are expected to attend and fully participate in all the scheduled classes, group learning opportunities, discussion sessions and practical sessions. In addition, punctuality is important. Students are expected to be on time and ready to begin at the stated time on the timetable.

Students who are not able to attend a formal contact session must provide the lecturer with a written note explaining their non-attendance. These notes will be kept in the course file in case queries regarding a student's poor performance are raised.

Misunderstandings about course content or failure to understand a topic can often be cleared up in one-on-one sessions with the lecturer and students are urged to consult their lecturers if they have any queries, etc. It is also important for a lecturer to be aware of any personal problems which may affect a student's performance.

Grievance procedures

Students who have grievances should first discuss the matter with the lecturer concerned. If reasonable cooperation from the lecturer is not forthcoming, the student may report it to the class representative, who will in turn take it to the SRC, who will then inform the Head of School who will then take the matter to the Registrar (in that order).

Academic dishonesty will NOT be tolerated in any form whatsoever. This includes cheating during any test, submitting someone else's work as your own (plagiarism) and copying someone else's program. Students caught cheating during a test (whether official or class test), will have their answer book/paper taken away, and will be handed a new answer book/paper to continue with. A committee consisting of the subject lecturers will then discuss the situation and make recommendations to the Programme Head. Where two or more students' handed-in programs seem undeniably to be identical copies of each other, ALL the students involved will receive a mark of ZERO.

Students with special needs

SPU is keen to accommodate students with special needs. These needs are catered for without influencing academic standards. It is, however, the student's responsibility to inform the lecturer in good time of any special needs to ensure effective communication in this regard. Learners are requested to make appointments to meet with lecturers during the available consultation times. As this will not always be possible, however, lecturers are also expected to make time to meet with students in 'emergency' situations.

Lecturers are expected to attend all scheduled contact classes. If possible, notice will be given if a lecturer cannot attend a class and arrangements will be made by the lecturer to cover the work missed. Students are advised to check the department notice board daily for any information pertaining to their course, classes, etc.

5. Student Contract: 2024 (Development Software)

a) Class timetables will be published on Moodle (http://learn.moodle.spu.ac.za)

b) Make sure you are registered as a student on Moodle for this subject.

c) A minimum mark of 50% is needed to pass this subject

d) Assignment due dates must be strictly adhered to.
e) A valid medical certificate must be handed in if absent from a test
f) Only ONE sick test will be written at the end of the year after the last test week.
g) Test mark queries will only be accommodated one week after marks discussion.
h) Plagiarism is both unethical and illegal and is regarded as a criminal offence.
i) Students are advised to attend all classes.
j) No games, distribution of pornography or hate speech allowed in classes & labs.
k) Academic dishonesty will NOT be tolerated in any form whatsoever.
l) Student must inform the lecturer in good time of any special needs.

This document is subject to revision. It is the responsibility of the student to ensure that the latest version of this document is used.

I hereby acknowledge that I have read and understood the Learner Guide for Development Software (NDEV74116) and I agree to the contents

Student Name: __________________________ **Number:** ___________________

Signature: ______________________________ **Date:** ______________________